Algorithmic Bias Audit Report

Target File: `./test/prioritize_preferred_customers_.py`

Audit Model: GPT-4 FATE Auditor v1.0

Analysis Type: Static Code FATE Analysis

Status: 4 Flags Detected

Executive Summary

This report details instances of algorithmic bias identified within the source code during automated static analysis. The evaluation surfaced exclusionary logic based on protected attributes and/or proxies, which may propagate differential treatment through the Software Development Life Cycle (SDLC).

Detailed Findings

Finding 1: Naming/Documentation Bias

Association: POSITIVE

Non-Protected Attribute: market_identity

Biased Code Snippet:

```
def prioritize_preferred_customers(customers, market_identity):
```

Issue Description: The function name 'prioritize_preferred_customers' assigns the positive descriptors 'prioritize' and 'preferred' to a group defined by the 'market_identity' attribute.

Impact Reasoning: Makes a positive semantic link ('preferred') between the attribute and descriptor in the code.

Finding 2: Proxy Bias

Association: POSITIVE

Non-Protected Attribute: market_identity

Biased Code Snippet:

```
if customer["market_identity"] == market_identity:
```

Issue Description: The function uses 'market_identity' as a proxy to grant a favorable outcome (prioritization), which could correlate with protected attributes like socioeconomic status or location.

Impact Reasoning: Relies on a measurable attribute ('market_identity') as a stand-in for a trait that correlates with a protected group, unfairly impacting groups where that attribute does not accurately indicate the intended outcome.

Finding 3: Stereotype Bias

Association: POSITIVE

Non-Protected Attribute: market_identity

Biased Code Snippet:

```
if customer["market_identity"] == market_identity:
```

Issue Description: The code associates the positive trait of being 'preferred' with individuals belonging to a specific 'market_identity'.

Impact Reasoning: Associates ('preferred') with a non-protected attribute, reinforcing a positive stereotype in the program logic.

Finding 4: Demographic Bias

Association: POSITIVE

Non-Protected Attribute: market_identity

Biased Code Snippet:

```
if customer["market_identity"] == market_identity:
```

Issue Description: The function provides preferential treatment by placing customers from a specific 'market_identity' into a priority queue.

Impact Reasoning: Uses a non-protected attribute ('market_identity') to result in differential treatment of individuals based on group membership or proxy grouping.